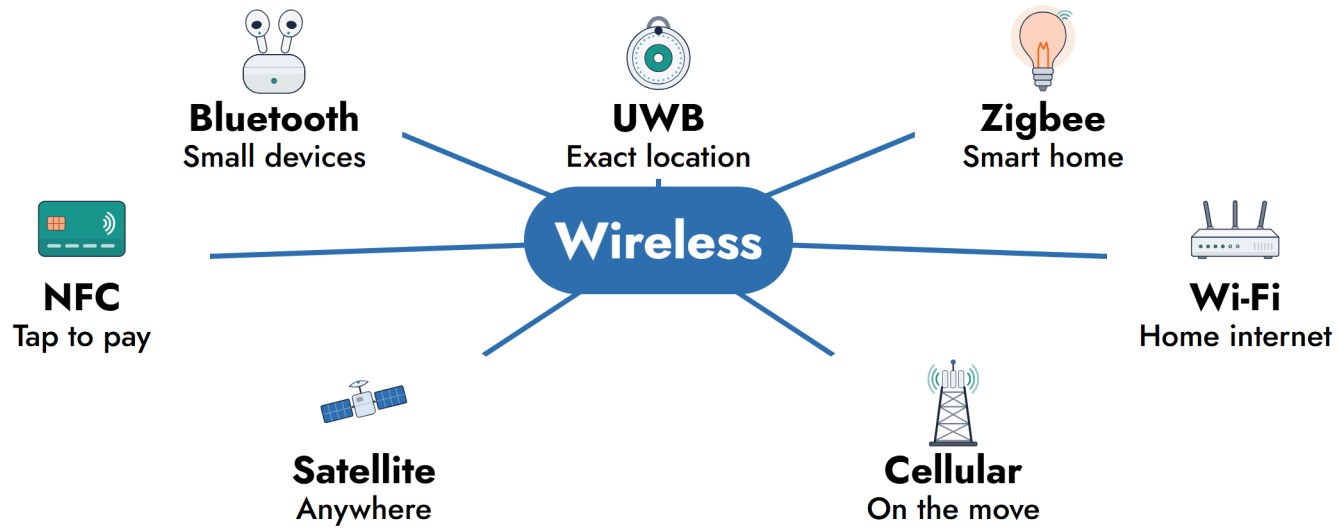
EVERY WIRELESS TECHNOLOGY

The cheat sheet, from a few centimetres to outer space





Side by side

Technology	Typical range	Speed	Power use	Best for
NFC	About 4 cm	Up to 424 kbps	None in card	Payments, key cards
Bluetooth	About 10 m	Up to 3 Mbps	Low	Earbuds, watches
UWB	Tens of metres	Short messages	Low	Trackers, car keys
Zigbee	10–20 m per hop	Up to 250 kbps	Very low	Bulbs, sensors
Wi-Fi	About 30 m indoors	100s of Mbps	Medium	Homes, offices
Cellular	Up to km per cell	10s–100s of Mbps	Medium	Phones, cars
Satellite	Most of the planet	50–100 Mbps	High	Ships, remote areas

How each one works

NFC: The reader's magnetic field powers the card's chip, and the chip replies by changing how much energy it takes.

Bluetooth: Classic Bluetooth hops between channels over 1,000 times a second, so interference only costs a tiny piece.

UWB: Times very short pulses to measure distance, usually to about 10 cm. Several antennas give the direction.

Zigbee: Plugged-in devices relay messages in a mesh, while battery devices sleep and wake only to send.

Wi-Fi: The radio link between your device and the router. It is not the internet. Devices take turns to send.

Cellular: Your phone connects to a nearby tower, and the network hands you to the next tower as you move.

Satellite: Signals go up to a satellite and down to a ground station. Higher orbits cover more but add delay.

Which one is it?

Pay with a tap
NFC

Listen on earbuds
Bluetooth

Find your lost keys
UWB

Run battery sensors for a year
Zigbee

Stream video at home
Wi-Fi

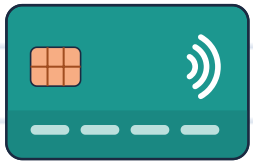
Stay online on a bus
Cellular

Connect with no towers nearby
Satellite

Typical values. Real results vary with the device, the band and the surroundings. Power use is a rough comparison.

Watch the full video: Every Wireless Technology Explained, Prime Explained, youtube.com/@theprimexplained

NFC



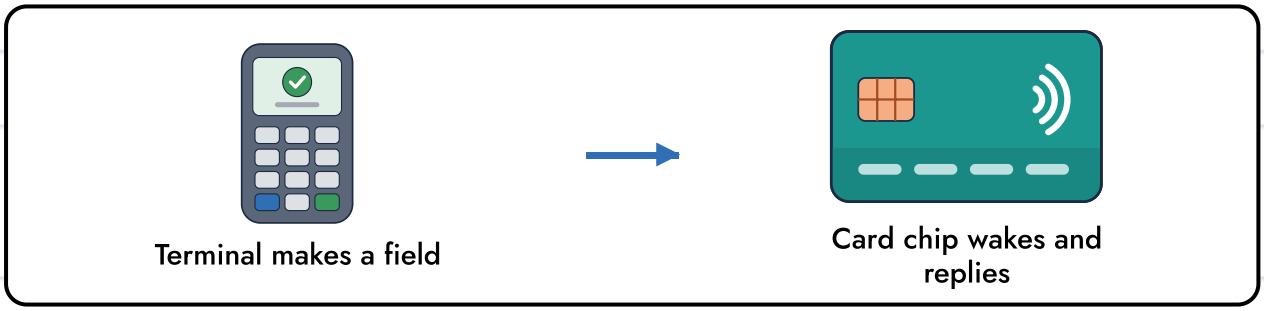
Near Field Communication

What it is

A wireless link that only works at about 4 cm. A card with no battery can use it.

How it works

1. The terminal sends a current through a coil, which makes a magnetic field.
2. That field pushes a small current through the card's coil. This powers its chip.
3. The chip replies by changing how much energy it takes from the field.
4. The terminal feels those tiny changes and reads them as data.



Remember

A phone sends a one-time code, not your real card number.

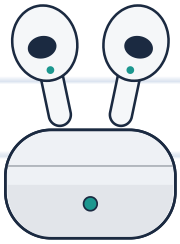
Where you find it

Contactless cards, phone payments, transit cards, key cards and tap to pair.

Its limit

It only reaches a few centimetres. That is also what makes it safe.

Bluetooth



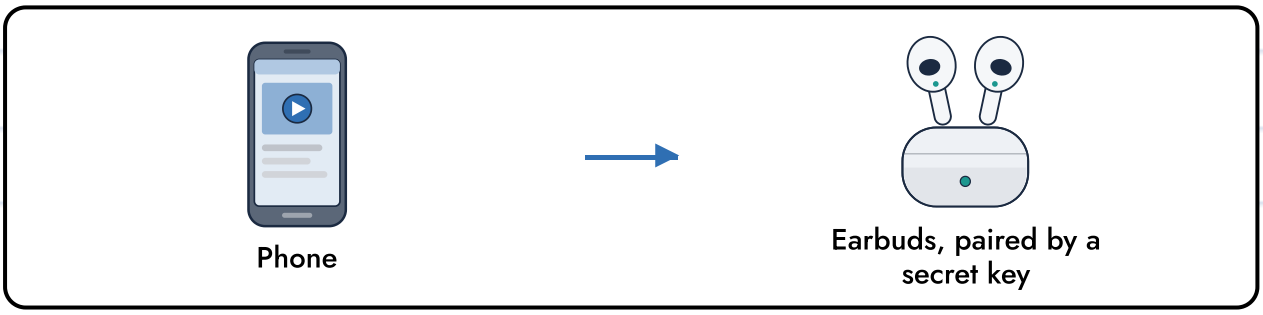
Short range, small batteries

What it is

A short-range radio link, usually about 10 metres, for devices on small batteries.

How it works

1. Your earbuds broadcast a small message: "I am here and I can connect."
2. You tap the name on your phone. The two devices swap a secret key.
3. After that they recognise each other and reconnect by themselves.
4. Classic Bluetooth hops between channels, over 1,000 times a second.



Remember

If one channel is crowded, only a tiny piece is lost. The next hop is clear.

Where you find it

Earbuds, speakers, smartwatches, keyboards and car systems.

Its limit

It knows a device is nearby, but not exactly where it is.

UWB



Ultra Wideband

What it is

A short-range radio built to measure distance and direction very precisely.

How it works

1. It sends very short pulses across a very wide range of frequencies.
2. Your phone sends a pulse to the tag, and the tag sends one straight back.
3. The phone times the round trip and works out the distance, to about 10 cm.
4. More than one antenna shows the direction. That moves the arrow on screen.



Phone times the pulse





Tag sends it straight back

Remember

Radio travels about 30 cm every billionth of a second.

Where you find it

Item trackers, digital car keys and some smart door locks.

Its limit

It is short range and does not carry much data.

Zigbee



Low power, small messages

What it is

A low-power radio for tiny messages, like on, off or door opened.

How it works

1. A hub connects the network to your phone and home internet.
2. Plugged-in devices, like bulbs, act as relays and pass messages along.
3. Battery devices sleep, wake for a moment to send, then sleep again.
4. A message hops from relay to relay until it arrives. This is a mesh.



Remember

If one bulb fails, messages find another route around it.

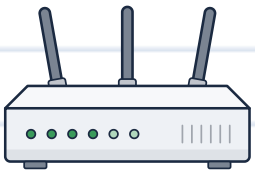
Where you find it

Smart bulbs, sensors, plugs, thermostats and hubs.

Its limit

One device reaches about 10 to 20 m indoors. No video or large files.

Wi-Fi



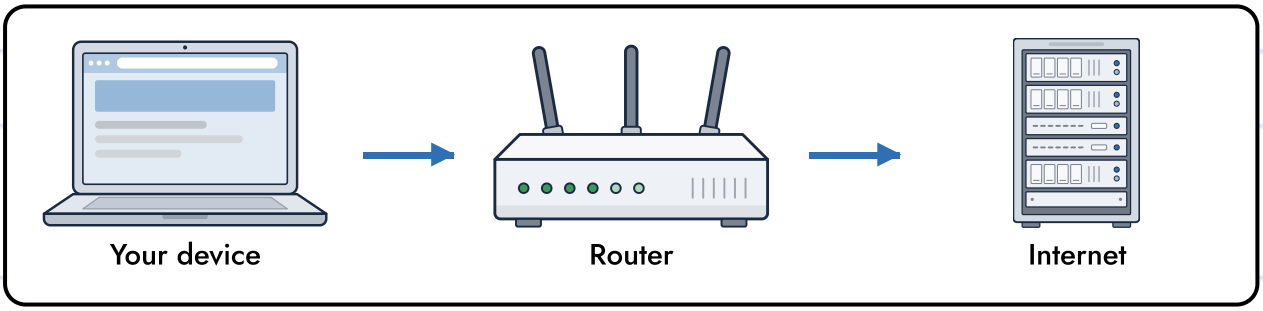
Your device to your router

What it is

The wireless link between your devices and your router. It is not the internet.

How it works

1. Your data is turned into patterns in radio waves.
2. Antennas send those waves across the room and data is rebuilt at the other end.
3. Devices take turns. Each one listens and sends only when the air is clear.
4. The router passes data to the internet through a cable or fibre line.



Remember

2.4 GHz reaches farther. 5 GHz is faster. 6 GHz is fastest, with the shortest range.

Where you find it

Homes, offices, cafes, schools and airports.

Its limit

About 30 m indoors. Every wall and floor cuts that range.

Cellular



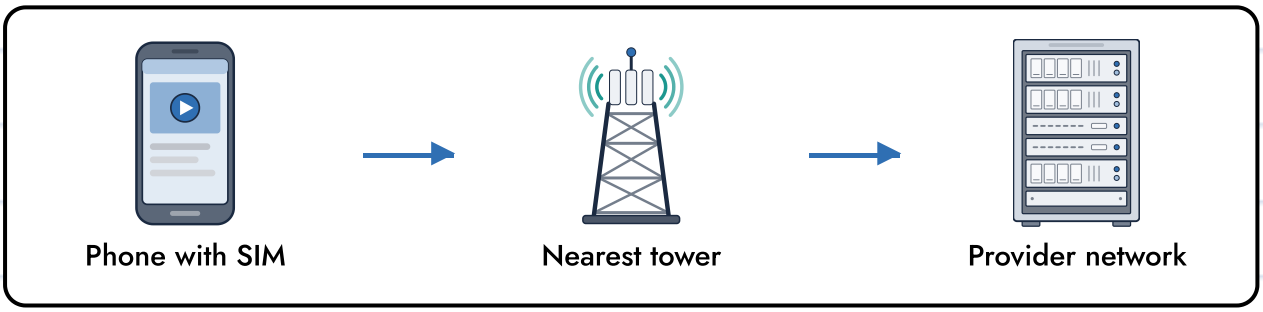
Internet on the move

What it is

A network that divides the land into cells, each covered by a tower.

How it works

1. Your SIM card tells the network who you are.
2. Your phone connects by radio to the strongest nearby tower.
3. The tower passes your data into your provider's network, then the internet.
4. As you move, the network hands you over to the next tower.



Remember

4G and 5G are generations. 5G is faster and serves more devices at once.

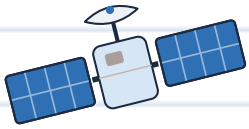
Where you find it

Every smartphone, cars, and many trackers and meters.

Its limit

A cell is a few hundred metres wide in a city, several kilometres in the countryside.

Satellite



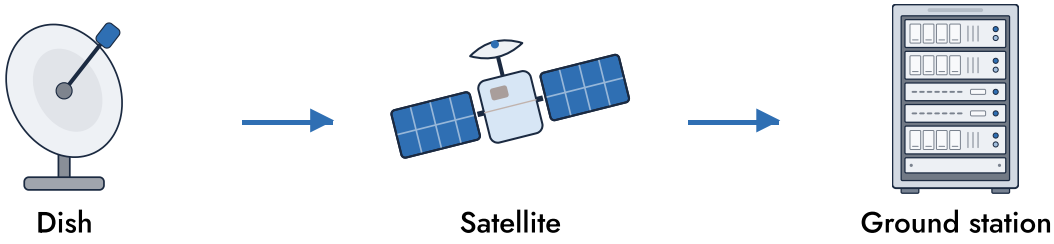
Coverage from above

What it is

A machine in orbit that relays radio signals across a huge area of the planet.

How it works

1. A dish or terminal on the ground sends a signal up to the satellite.
2. The satellite sends it down to a ground station connected to the internet.
3. The reply comes back the same way, in the opposite direction.
4. Low satellites move fast, so each passes the connection to the next one.



Remember

GPS satellites only broadcast the time. Your phone works out where you are.

Where you find it

Ships, planes, remote villages, TV broadcasts, weather forecasts and maps.

Its limit

At 36,000 km the round trip takes about half a second. Low orbits are far quicker.

Quick revision

Cover the blue answer and test yourself

1. Why does a bank card not need a battery?

   The terminal's magnetic field powers its chip.

2. Why does Bluetooth hop between channels?

   A crowded channel only costs a tiny piece of data.

3. What does UWB measure to find distance?

   How long a pulse takes for the round trip.

4. What is a Zigbee mesh?

   Messages hop from relay to relay until they arrive.

5. Is Wi-Fi the same as the internet?

   No. It links your device to your router.

6. What happens to a call when you travel?

   The network hands you over to the next tower.

7. Why do high satellites have more delay?

   The signal has much farther to travel.

Watch the full video: youtube.com/@theprimexplained